Broadcast Traffic Control using SDN

Output screenshots:

```
dhriti@dhriti27:~/pox$ ./pox.py broadcast_control
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
WARNING:version:POX requires one of the following versions of Python: 3.6 3.7 3.8 3.9
WARNING:version:You're running Python 3.12.
WARNING:version:If you run into problems, try using a supported version.
INFO:core:POX 0.7.0 (gar) is up.
INFO:openflow.of_01:[00-00-00-00-00-01 1] connected
INFO:openflow.of_01:[00-00-00-00-00-01 1] closed
INFO:openflow.of_01:[00-00-00-00-00-01 3] connected
INFO:broadcast_control:Broadcast packet count=1
INFO:broadcast_control:Broadcast packet count=2
INFO:broadcast_control:Broadcast packet count=3
INFO:broadcast_control:🚫 Blocking ALL broadcast traffic
INFO:openflow.of_01:[00-00-00-00-00-01 3] closed
```

```
dhriti@dhriti27:~/pox$ sudo mn --topo single,4 --controller remote
*** Creating network
*** Adding controller
Unable to contact the remote controller at 127.0.0.1:6653
Connecting to remote controller at 127.0.0.1:6633
*** Adding hosts:
h1 h2 h3 h4
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1) (h3, s1) (h4, s1)
*** Configuring hosts
h1 h2 h3 h4
*** Starting controller
c0
*** Starting 1 switches
s1 ...
```

```
*** Starting CLI:
mininet> h1 ping h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=6.27 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=2.47 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=2.33 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=2.67 ms
64 bytes from 10.0.0.2: icmp_seq=5 ttl=64 time=2.52 ms
^C
--- 10.0.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4016ms
rtt min/avg/max/mdev = 2.328/3.250/6.269/1.512 ms
mininet> h1 ping h3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=3.77 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=2.58 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=2.56 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=1.53 ms
^C
--- 10.0.0.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 1.531/2.608/3.773/0.793 ms
```

```
*** Starting CLI:
mininet> h1 iperf -s &
mininet> h2 iperf -c 10.0.1
------------------------------------------------------------
Client connecting to 10.0.1, TCP port 5001
TCP window size: 85.3 KByte (default)
------------------------------------------------------------
[  1] local 10.0.0.2 port 40152 connected with 10.0.0.1 port 5001 (icwnd/mss/irtt=14/1448/3187)
[ ID] Interval       Transfer     Bandwidth
[  1] 0.0000-10.3266 sec  81.9 MBytes  66.5 Mbits/sec
```

```
dhriti@dhriti27:~/pox$ sudo ovs-ofctl dump-flows s1
 cookie=0x0, duration=79.243s, table=0, n_packets=9, n_bytes=378, priority=100,dl_dst=ff:ff:ff:ff:ff:ff actions=drop
```